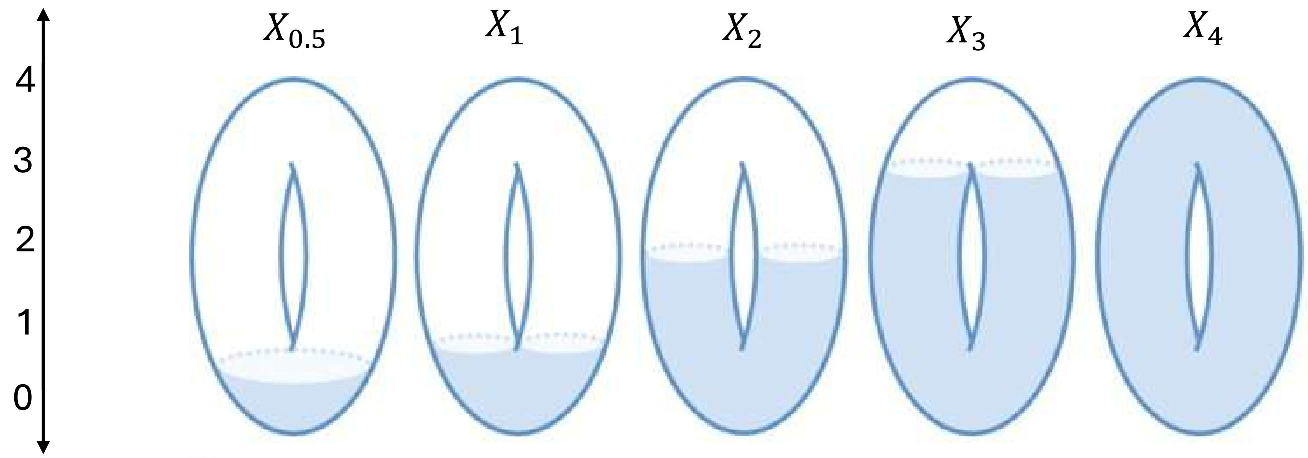
AATRN Tutorial-a-thon 2025

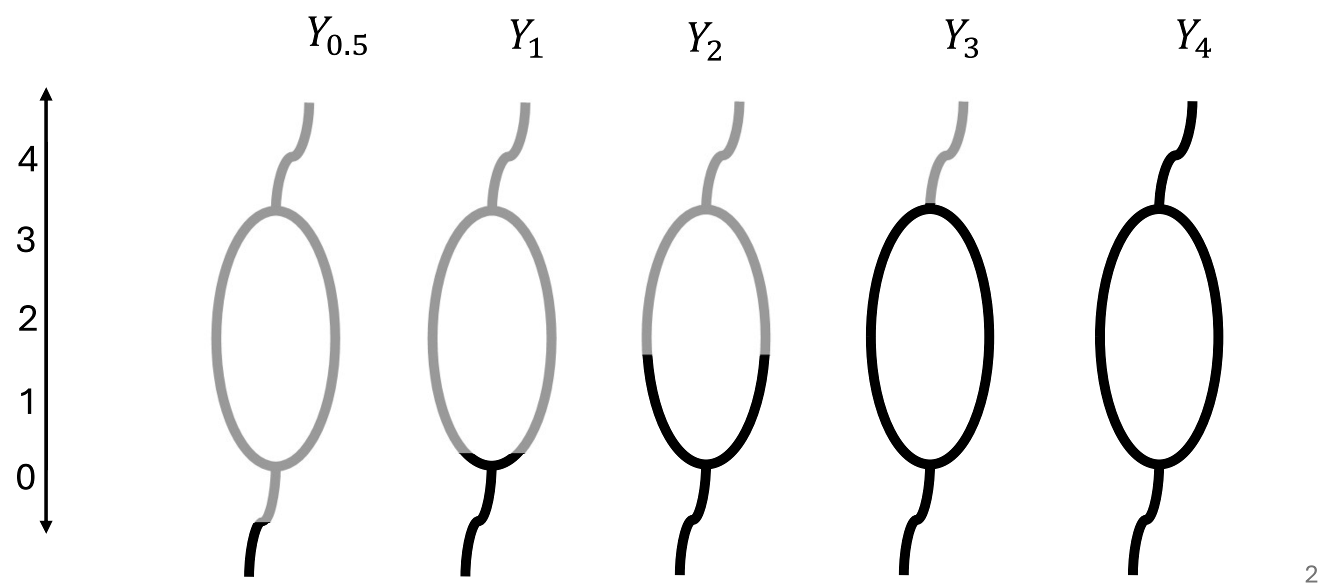
Generalizing the interleaving distance using categorical flows

Astrid A. Olave H.
olaveher@msu.edu

Let us start comparing two persistence modules



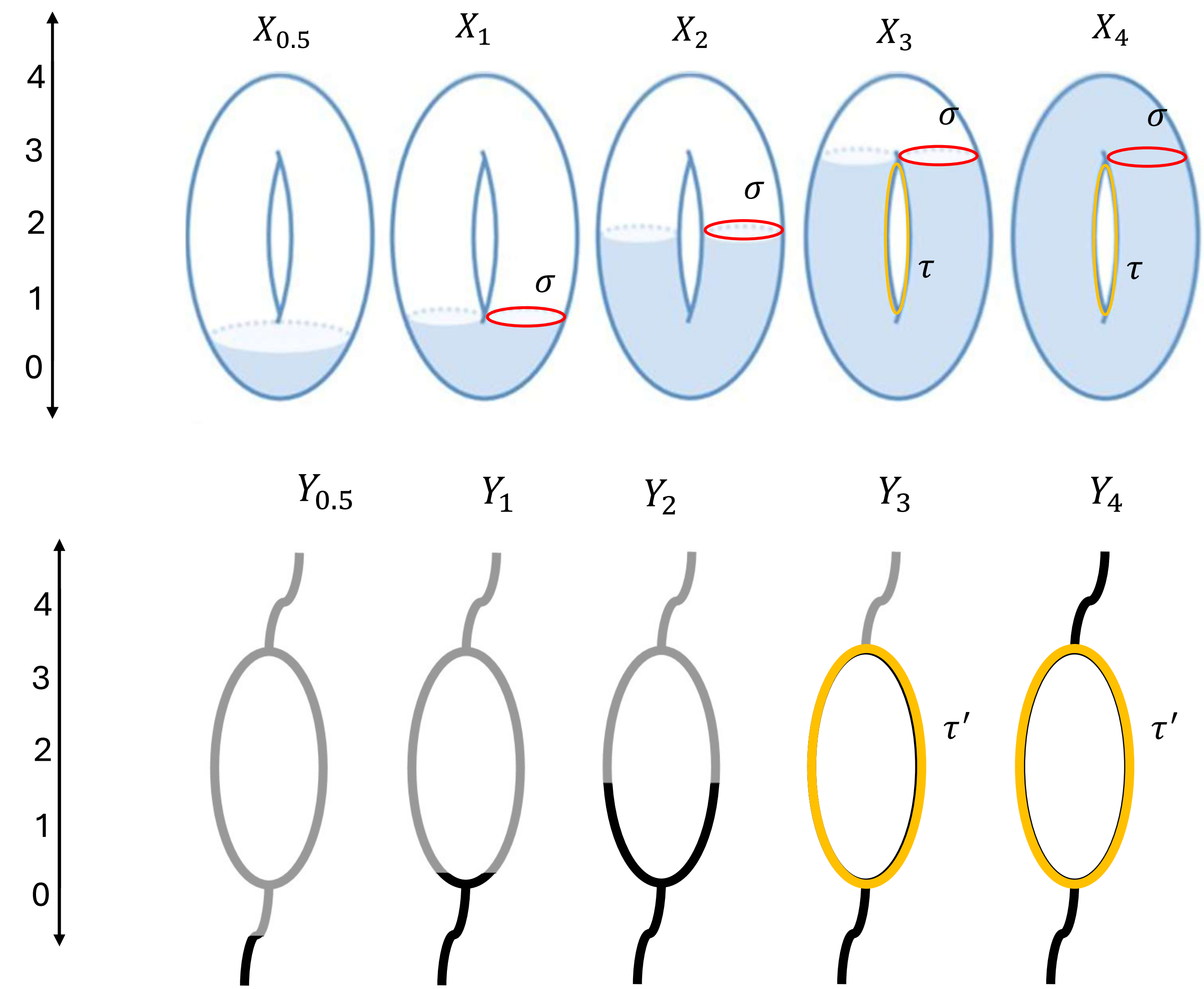
1.



1. Image modified from : Curry, J. M. (2015). Topological data analysis and cosheaves. *Japan Journal of Industrial and Applied Mathematics*, *32*(2), 333–371. https://doi.org/10.1007/s13160-015-0173-9

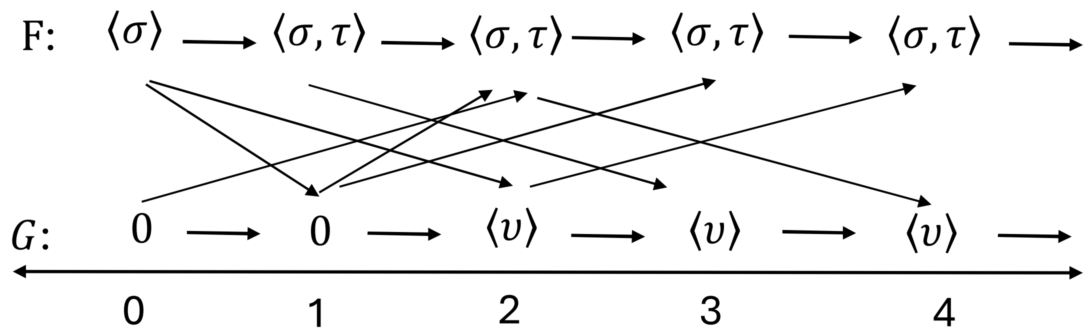
Let us start comparing two persistence modules

using homology



1. Image modified from : Curry, J. M. (2015). Topological data analysis and cosheaves. *Japan Journal of Industrial and Applied Mathematics*, *32*(2), 333–371. https://doi.org/10.1007/s13160-015-0173-9

When two persistent modules are ϵ-interleaved ?



For $\epsilon > 0$, we say that F and G **are ϵ-interleaved** if there is a pair of natural transformations φ and ψ such that for all $a \in \mathbb{R}$ the diagrams commute .

$$\begin{tikzcd}
F(a) \arrow[r] \arrow[d, "\varphi(a)"'] & F(b) \arrow[d, "\varphi(b)"] \\
G(a+\varepsilon) \arrow[r] & G(b+\varepsilon)
\end{tikzcd}
\qquad
\begin{tikzcd}
F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(a+2\varepsilon) \\
& G(a+\varepsilon) \arrow[ur, "\psi(a+\varepsilon)"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
F(a+\varepsilon) \arrow[r] & F(b+\varepsilon) \\
G(a) \arrow[r] \arrow[u, "\psi(a)"] & G(b) \arrow[u, "\psi(b)"']
\end{tikzcd}
\qquad
\begin{tikzcd}
& F(a+\varepsilon) \arrow[dr, "\varphi(a+\varepsilon)"] & \\
G(a) \arrow[rr] \arrow[ur, "\psi(a)"] & & G(a+2\varepsilon)
\end{tikzcd}$$

$$
\begin{tikzcd}
\langle \sigma \rangle \arrow[rr] \arrow[dr] & & \langle \sigma, \tau \rangle \\
& 0 \arrow[ur] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\langle \sigma, \tau \rangle \arrow[rr] \arrow[dr] & & \langle \sigma, \tau \rangle \\
& \langle \upsilon \rangle \arrow[ur] &
\end{tikzcd}
$$

F and G are not ϵ-interleaved for any ϵ.
The triangle diagrams do not commute.

$$
\begin{tikzcd}
\mathrm{F:} & \langle \sigma \rangle \arrow[r] \arrow[drr] & \langle \sigma, \tau \rangle \arrow[r] \arrow[drr] & \langle \sigma, \tau \rangle \arrow[r] \arrow[drr] & \langle \sigma, \tau \rangle \arrow[r] & \langle \sigma, \tau \rangle \\
G: & 0 \arrow[r] \arrow[urr] & 0 \arrow[r] \arrow[urr] & \langle \upsilon \rangle \arrow[r] \arrow[urr] & \langle \upsilon \rangle \arrow[r] & \langle \upsilon \rangle \\
& 0 & 1 & 2 & 3 & 4
\end{tikzcd}
$$

$$d_I(F, G) \coloneqq \infty$$

What is the interleaving distance between modules?

We define ***the interleaving distance*** between the modules F and G by

$$d_I(F, G) \coloneqq \inf\{\epsilon \geq 0 \mid F, G \text{ are } \epsilon\text{−interleaved}\}$$

Redefining our terms using categorical language

In 2014, Bubenik and Scott redevelop topological persistence from a categorical point of view.

A **persistent module** F is a functor from $\mathbb{R}$ (poset) to Vec. (Actually, any category $\boldsymbol{C}$)

An ***ϵ-interleaving*** of F and G consists of natural transformations

$$\varphi : F \Rightarrow G\mathcal{T}_\epsilon \text{ and } \psi : G \Rightarrow F\Omega_\epsilon$$

$$\begin{tikzcd} F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(b) \arrow[dr, "\varphi(b)"] & \\ & G(a+\varepsilon) \arrow[rr] & & G(b+\varepsilon) \end{tikzcd} \qquad \begin{tikzcd} & F(a+\varepsilon) \arrow[rr] & & F(b+\varepsilon) \\ G(a) \arrow[rr] \arrow[ur, "\psi(a)"] & & G(b) \arrow[ur, "\psi(b)"'] & \end{tikzcd}$$

such that

$$(\psi\Omega_\epsilon)\varphi \;=\; F\eta_{2\epsilon} \text{ and } (\varphi\Omega_\epsilon)\psi \;=\; G\eta_{2\epsilon}$$

$$\begin{tikzcd} F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(a+2\varepsilon) \\ & G(a+\varepsilon) \arrow[ur, "\psi(a+\varepsilon)"'] & \end{tikzcd} \qquad \begin{tikzcd} & F(a+\varepsilon) \arrow[dr, "\varphi(a+\varepsilon)"] & \\ G(a) \arrow[rr] \arrow[ur, "\psi(a)"] & & G(a+2\varepsilon) \end{tikzcd}$$

For $b \geq 0$, define $\Omega_b : (\mathbb{R}, \leq) \rightarrow (\mathbb{R}, \leq)$ to be the shift functor given by $\Omega_b(a) = a + b$,
and define $\eta_b : Id(\mathbb{R}, \leq) \Rightarrow \Omega_b$ to be the natural transformation given by $\eta_b(a) : a \leq a + b$.

Categories open a new world for persistence modules

A *persistence module* is a functor F: $P \to C$ with P is a preorder set and C any category (?)

P = zig-zag

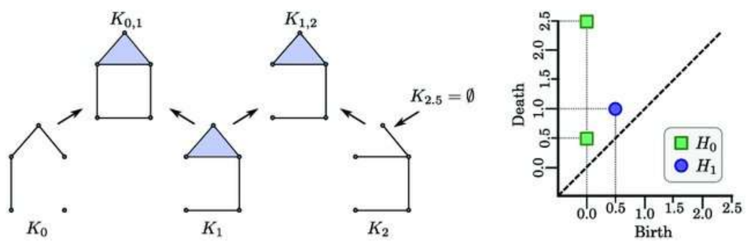


Image from Myers, A., Muñoz, D., Khasawneh, F. A., & Munch, E. (2023). Temporal network analysis using zigzag persistence. *EPJ Data Science*, *12*(1).

$P = \mathbb{R}^n$

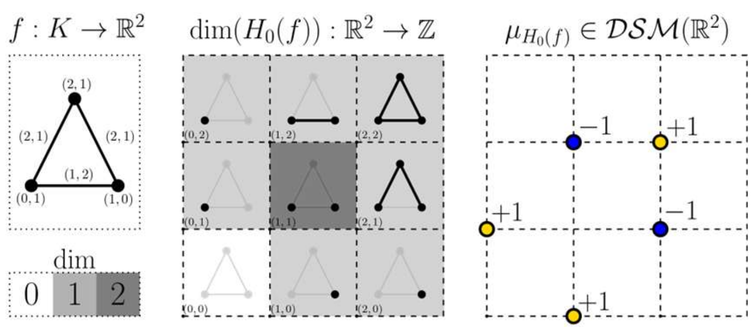


Image from Scoccola, L., Setlur, S., Loiseaux, D., Carrière, M., & Oudot, S. (2024). Differentiability and Optimization of Multiparameter Persistent Homology. *arXiv preprint arXiv:2406.07224*.

- C is essentially small symmetric monoidal category with images. e.g $FinSet, Vec, Ab, FinAb, Rep(\mathbb{N})$
- C is also bicomplete
- C is a Grothendieck category

  e.g $Rmod, Vec$

P is lattice

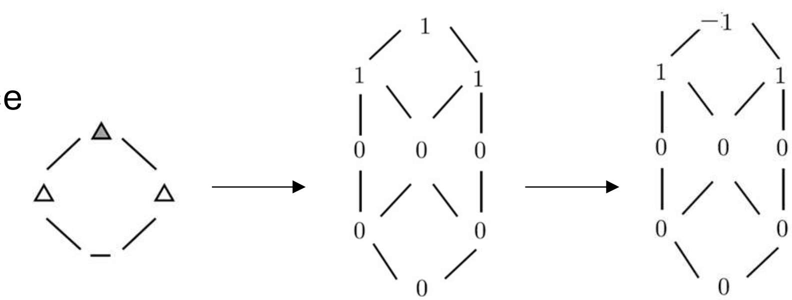


Image modified from McCleary, A., & Patel, A. (2022). Edit distance and persistence diagrams over lattices. *SIAM Journal on Applied Algebra and Geometry*, *6*(2), 134-155.

Interleaving distance between objects instead of functors

In 2018, De Silva, Munch, and Stefanou generalize interleavings defining categorical flows.

Recall this definitions to define interleaving distance

For $b \geq 0$, define $\Omega_b : (\mathbb{R}, \leq) \to (\mathbb{R}, \leq)$ to be the functor given by $\Omega_b(a) = a + b$, and
Define $\eta_b : Id(\mathbb{R}, \leq) \Rightarrow T_b$ to be the natural transformation given by $\eta_b(a) : a \leq a + b$.

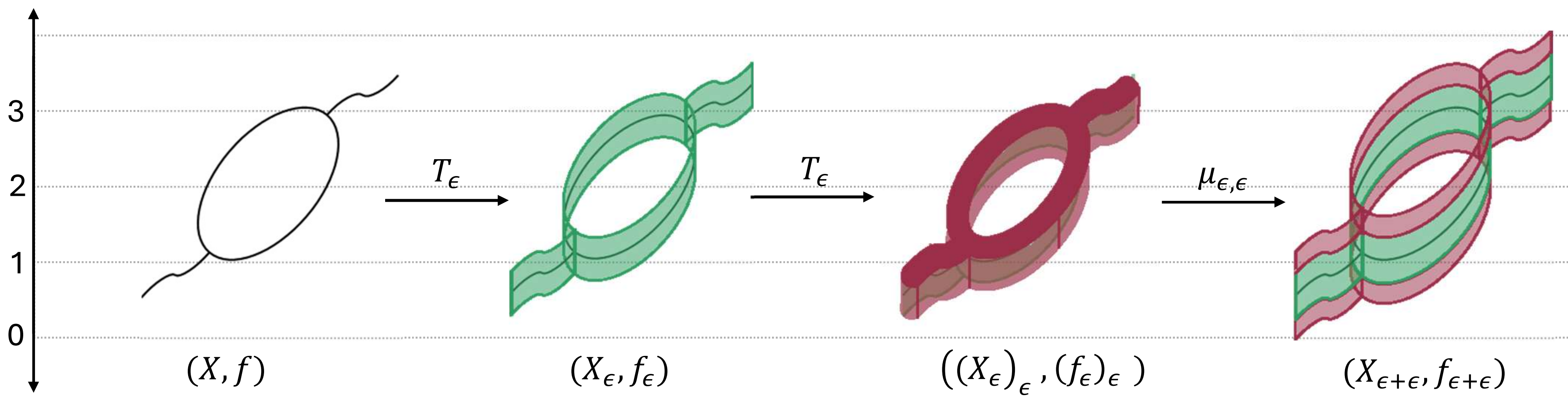
2.3. DEFINITION. *A coherent* $[0, \infty)$*-action* $\mathcal{T} = (\mathcal{T}, u, \mu)$ *on a category* $\mathcal{C}$ *is said to be a* ***flow***. *Specifically, a flow* $\mathcal{T}$ *on a category* $\mathcal{C}$ *consists of*

- *a functor* $\mathcal{T} : ([0, \infty), \leq) \to \mathbf{End}(\mathcal{C})$, $\varepsilon \mapsto \mathcal{T}_\varepsilon$,
- *a natural transformation* $u : I_\mathcal{C} \Rightarrow \mathcal{T}_0$, *where* $I_\mathcal{C}$ *is the identity endofunctor of* $\mathcal{C}$, *and*
- *a collection of natural transformations* $\mu_{\varepsilon,\zeta} : \mathcal{T}_\varepsilon \mathcal{T}_\zeta \Rightarrow \mathcal{T}_{\varepsilon+\zeta}$, $\varepsilon, \zeta \geq 0$,

An example of flows using $\mathbb{R}$-spaces

Consider $(Top \downarrow \mathbb{R})$ the category of real valued functions $f: X \to \mathbb{R}$, noted as the pair (X, f).

Let us define a flow on $(Top \downarrow \mathbb{R})$ as follows.

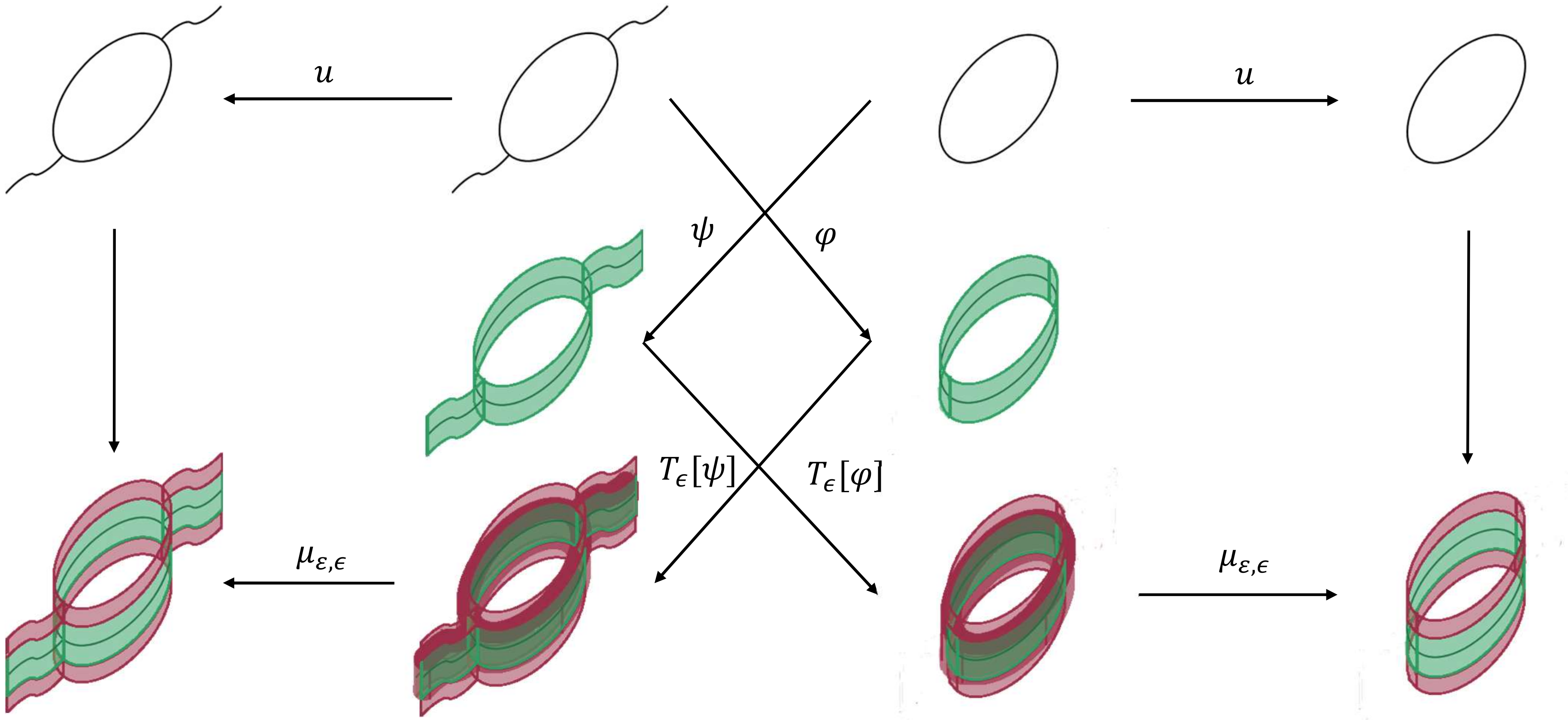


- The functor $T_\varepsilon: (Top \downarrow \mathbb{R}) \to (Top \downarrow \mathbb{R})$ given by

$$T_\varepsilon(X, f) = (X_\varepsilon, f_\varepsilon) \quad \text{where} \quad X_\varepsilon \coloneqq X \times [-\varepsilon, \varepsilon] \ \text{ and } f_\varepsilon(x, s) = f(x) + s$$

- $u: (X, f) \to (X_0, f_0)$
- $\mu_{\varepsilon,\gamma}: \left((X_\gamma)_\varepsilon, (f_\gamma)_\varepsilon\right) \to (X_{\varepsilon+\gamma}, f_{\varepsilon+\gamma})$

When two objects are ϵ-interleaved ?

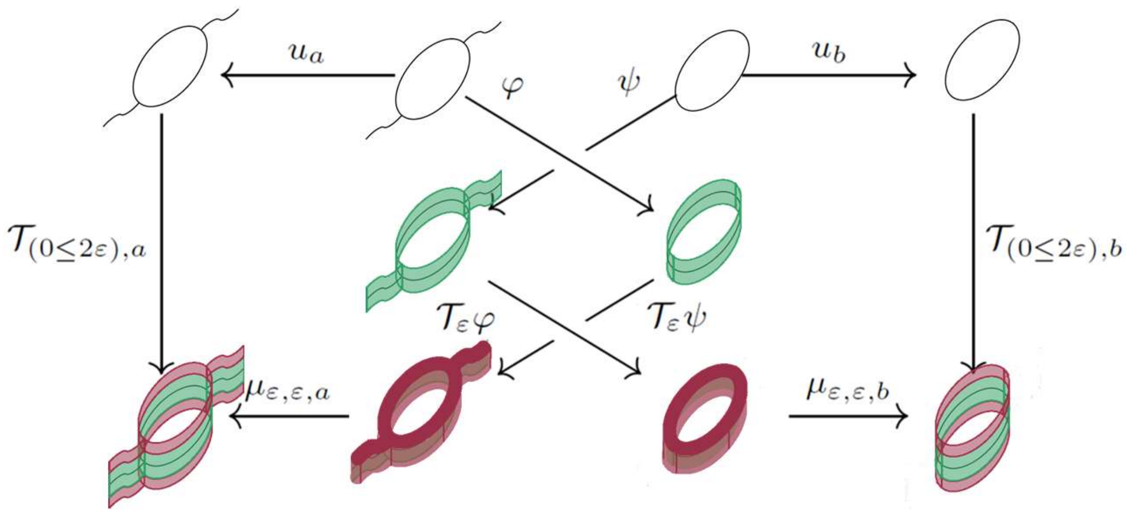


We say that (X, f) and (Y, g) **are weakly ϵ-interleaved** if there is a pair of morphisms φ and ψ such that the diagrams commute .

What is the interleaving distance between objects ?

The interleaving distance with respect to T for a pair of objects a, b in C is defined to be

$$d_{(C,T)}(a, b) = \inf\{\varepsilon \geq 0 \mid a, b \text{ are weakly } \varepsilon\text{−interleaved}\}$$



Theorem: The interleaving distance on $(Top \downarrow \mathbb{R})$ induced by the flow , coincides with the L^∞-distance for real valued functions. That is, for any $f: X \to \mathbb{R}$, $g: Y \to \mathbb{R}$

$$d_{((Top \downarrow \mathbb{R}),\mathrm{T})}(f, g) = \inf_{\phi: X \to Y} \|f - g \circ \phi\|_\infty$$

Distances coming from interleavings induced by flows

l^∞ cophenetic distance for phylogenetic trees

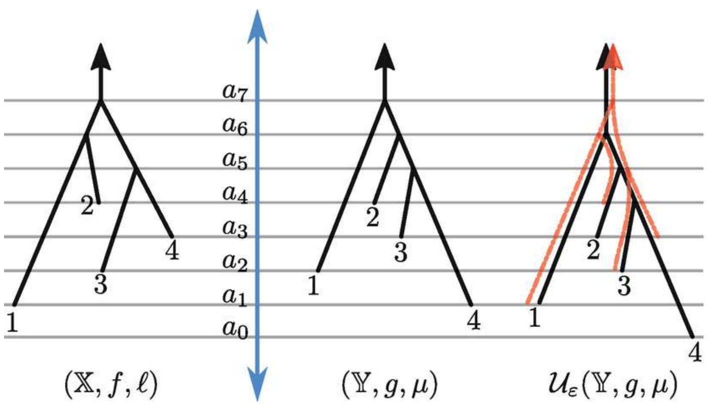


Image modified from: Munch, E., & Stefanou, A. (2019). The ℓ∞-cophenetic metric for phylogenetic trees as an interleaving distance. In *Research in Data Science* (pp. 109-127). Cham: Springer International Publishing.

Distance between (labeled) Reeb graphs

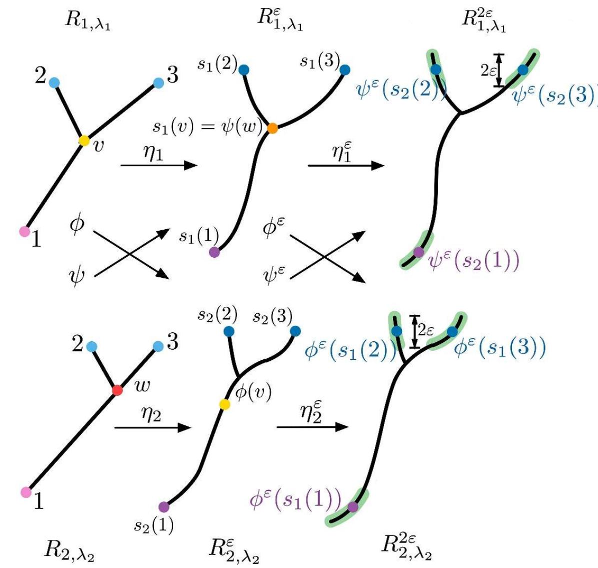
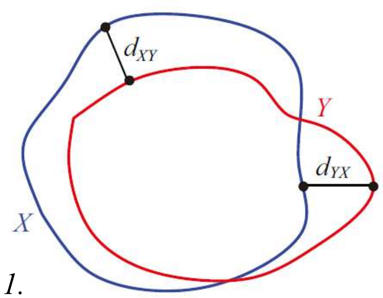


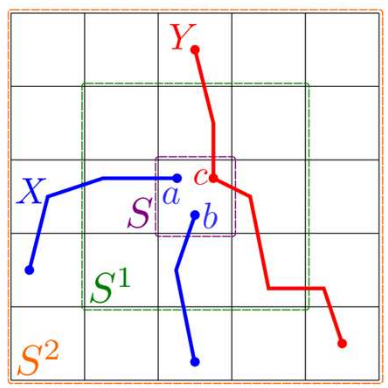
Image modified from: Lan, F., Parsa, S., & Wang, B. (2024). Labeled interleaving distance for Reeb graphs. *Journal of Applied and Computational Topology*, *8*(8), 2367-2399.

Hausdorff distance



1.

Distance Between Mapper graphs



2.

1. Image retrieved from: Grand Challenge. (2019). *Evaluation Criteria*. STRUCTSEG2019 . https://structseg2019.grand-challenge.org/Evaluation/

2. Chambers, E., Munch, E., Percival, S., & Wang, B. (2023). Bounding the Interleaving Distance for Mapper Graphs with a Loss Function. *arXiv preprint arXiv:2307.15130*.